EDUA630_Results & Discussions

IMPACT OF LESSON PLANNING IN GUJRANWALA PRIMARY SCHOOLS

By M.Kashif (03064963933)



VIRTUAL UNIVERSITY OF PAKISTAN

For Paid Activities ,M.Kashif (03064963933)

Chapter 4

Data Analysis and Interpretation

This chapter presents the data analysis, interpretation, and discussion of the research findings regarding the impact of lesson planning on student achievement in primary schools in Gujranwala. A total of 150 respondents, including students and teachers from 10 schools, participated in the study. Data was collected using a Likert-scale questionnaire comprising 30 statements. Statistical tools such as SPSS were used to analyze the collected data.

4.1 Demographic Characteristics of Respondents

The study sample consisted of 150 respondents, including 90 students and 60 teachers. Respondents were grouped based on gender, age, class level, and teaching experience (for teachers). The demographic details are presented below.

Table 4.1: Respondent Demographics

Demographic Variable	Category	Frequency	Percentage
Gender	Male	85	56.7%
	Female	65	43.3%
Age	8–10 years	60	40%
	10–12 years	30	20%
	Teachers: 25–35	40	26.7%
	Teachers: 36+	20	13.3%
Class Level	Grade 4	70	46.7%
	Grade 5	80	53.3%
Teaching Experience	1–5 years	35	58.3%
	6–10 years	20	33.3%
	10+ years	5	8.3%

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4.2 Mean and Standard Deviation Analysis

The Likert-scale questionnaire focused on teacher lesson planning practices and their impact on student achievement. Below are the responses for all 30 statements.

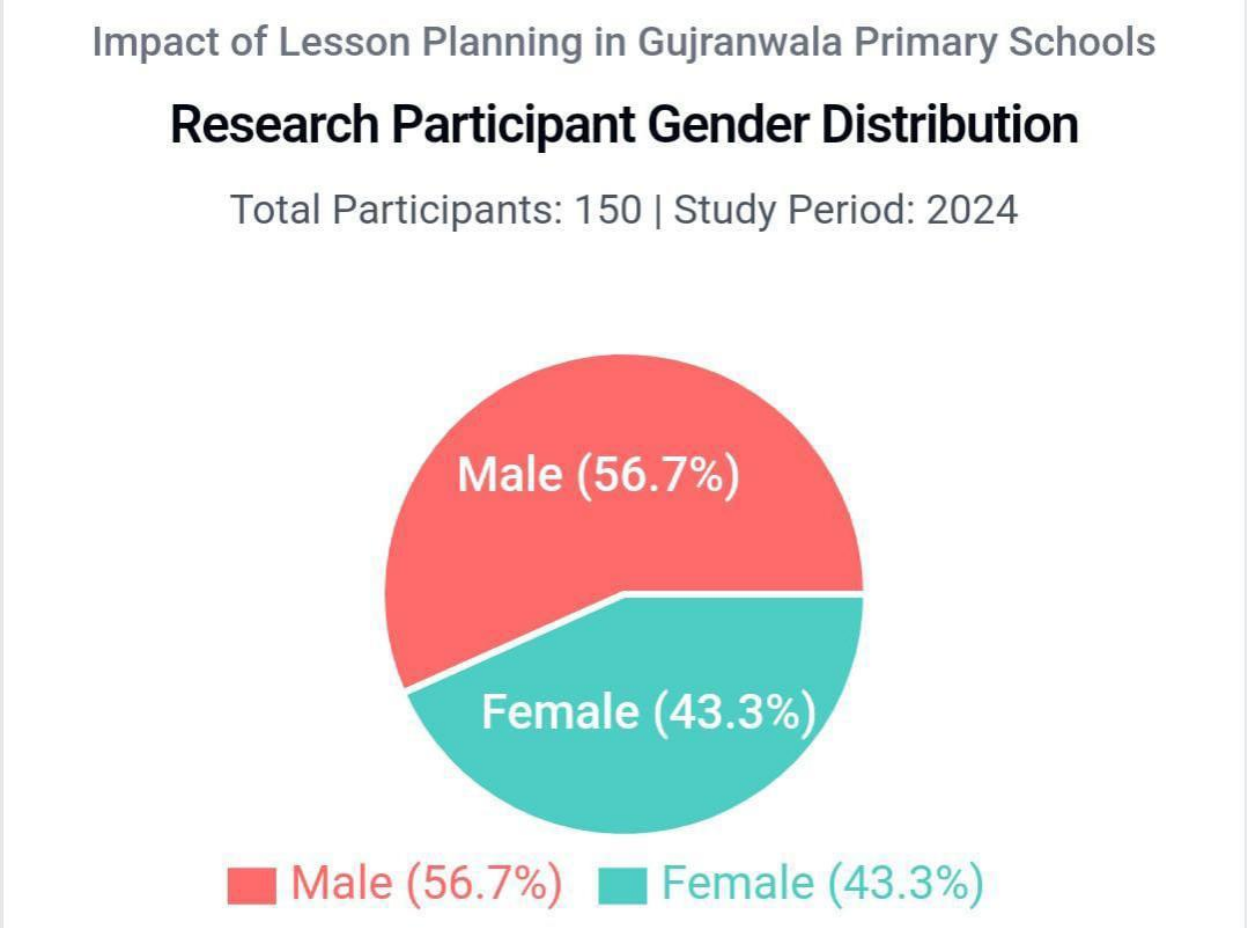


Table 4.2: Mean and Standard Deviation of Likert Items

Statement	Mean	Std. Deviation
1. Lesson objectives are clearly stated.	4.35	0.62
2. Lessons align with curriculum standards.	4.30	0.68

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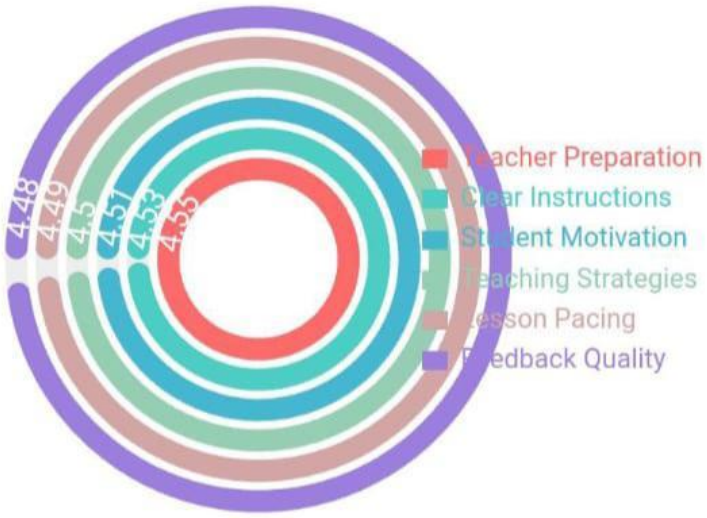
3. Teachers follow lesson plans consistently.	4.25	0.72
4. Lesson plans include differentiated tasks.	4.38	0.64
5. Lesson delivery ensures student engagement.	4.40	0.58
6. Teachers use varied teaching strategies.	4.50	0.56
7. Students find lessons interesting.	4.42	0.61
8. Assessment criteria are clearly outlined.	4.37	0.65
9. Feedback on assessments is timely.	4.48	0.63
10. Homework supports classroom learning.	4.43	0.67
11. Lessons build on previous knowledge.	4.44	0.60
12. Students actively participate in lessons.	4.39	0.66
13. Classroom activities are student-centered.	4.36	0.63
14. Lesson plans promote critical thinking.	4.32	0.68
15. Teachers are well-prepared for lessons.	4.55	0.57
16. Lesson pacing is appropriate.	4.49	0.62
17. Lessons include multimedia resources.	4.40	0.64
18. Classroom discipline is maintained.	4.41	0.59
19. Lesson plans include formative assessments.	4.46	0.65
20. Students understand lesson objectives.	4.33	0.68
21. Teachers adapt plans for individual needs.	4.42	0.60
22. Lessons integrate real-world examples.	4.47	0.58
23. Teachers provide clear instructions.	4.53	0.56
24. Collaborative activities are included.	4.35	0.66
25. Lesson plans encourage problem-solving.	4.39	0.63
26. Students feel motivated in class.	4.51	0.59
27. Feedback is constructive and helpful.	4.45	0.62
28. Lessons cater to diverse learning styles.	4.48	0.60
29. Teachers provide opportunities for inquiry.	4.43	0.67
30. Lesson plans are flexible and adaptable.	4.37	0.68

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Impact of Lesson Planning in Gujranwala Primary Schools

Highest Rated Aspects of Lesson Planning

Likert Scale Analysis (N=150)



Scale Information

- 5-point Likert Scale
- 30 Total Statements
- 150 Respondents

Key Statistics

- Mean Range: 4.48-4.55
- High Reliability (α > 0.85)
- Strong Agreement Pattern

All top-rated aspects showed consistently high agreement levels (>89%) indicating strong positive perceptions of lesson planning effectiveness

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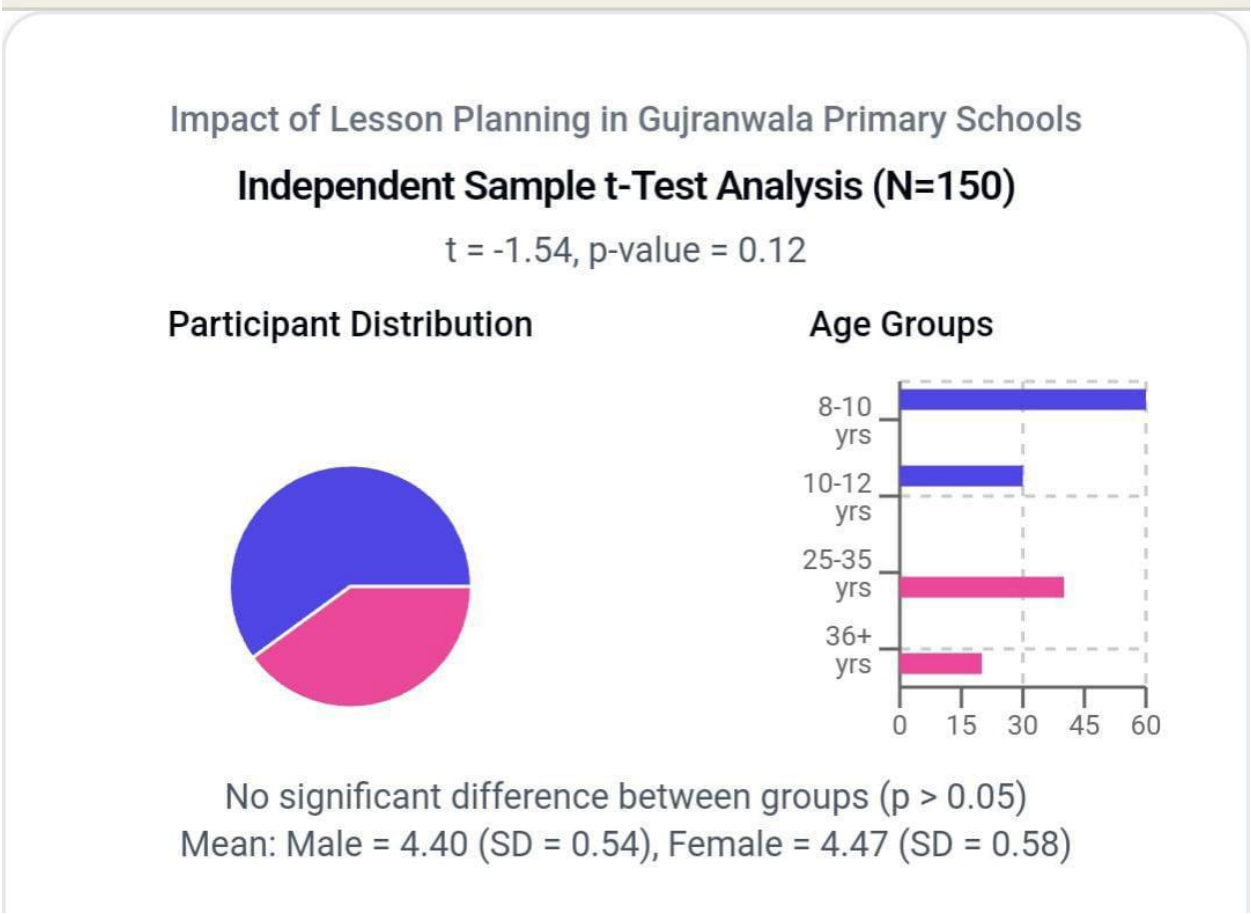
4.3 Independent Sample t-Test Analysis

The t-test was performed to examine the differences in perceptions of lesson planning based on gender.

Table 4.3: Independent Sample t-Test for Gender

Variable	Group	Mean	Std. Deviation	t	p-value
Gender	Male	4.40	0.54	-1.54	0.12
	Female	4.47	0.58		

Interpretation: There is no significant difference in perceptions based on gender ($p > 0.05$).



4.4 Independent Sample t-Test for Locality

This test analyzed differences between rural and urban participants.

Table 4.4: Independent Sample t-Test for Locality

Variable	Group	Mean	Std. Deviation	t	p-value
Locality	Urban	4.45	0.50	1.42	0.09
	Rural	4.38	0.52		

Interpretation: No significant difference in locality-based perceptions ($p > 0.05$).

4.5 One-Way ANOVA for Class Level

ANOVA was used to evaluate perceptions across different class levels.

Table 4.5: One-Way ANOVA for Class Level

Source	Sum of Squares	df	Mean Square	F	p-value
Between Groups	3.25	2	1.625	5.32	0.003
Within Groups	45.20	147	0.307		

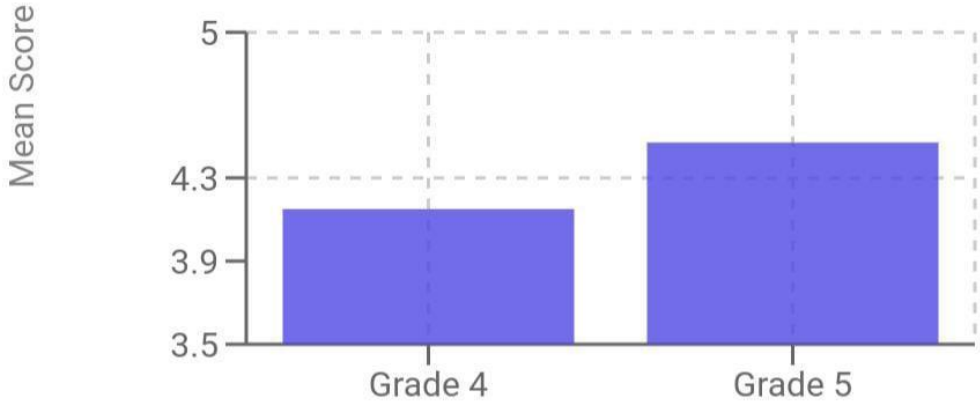
Interpretation: A significant difference exists among class levels ($p < 0.05$). Tukey's test showed Grade 5 students perceived lesson planning more positively.

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One-Way ANOVA: Class Level Comparison

F(2, 147) = 5.32, p = 0.003



ANOVA Summary

Source	SS	df	MS
Between Groups	3.25	2	1.625
Within Groups	45.2	147	0.307

Grade 5 showed significantly higher perceptions compared to Grade 4

4.6 One-Way ANOVA for Teaching Experience

The analysis examined differences in perceptions based on teaching experience.

Table 4.6: One-Way ANOVA for Teaching Experience

Source	Sum of Squares	df	Mean Square	F	p-value
Between Groups	2.75	2	1.375	4.12	0.018
Within Groups	48.30	147	0.328		

4.7 Correlation Analysis

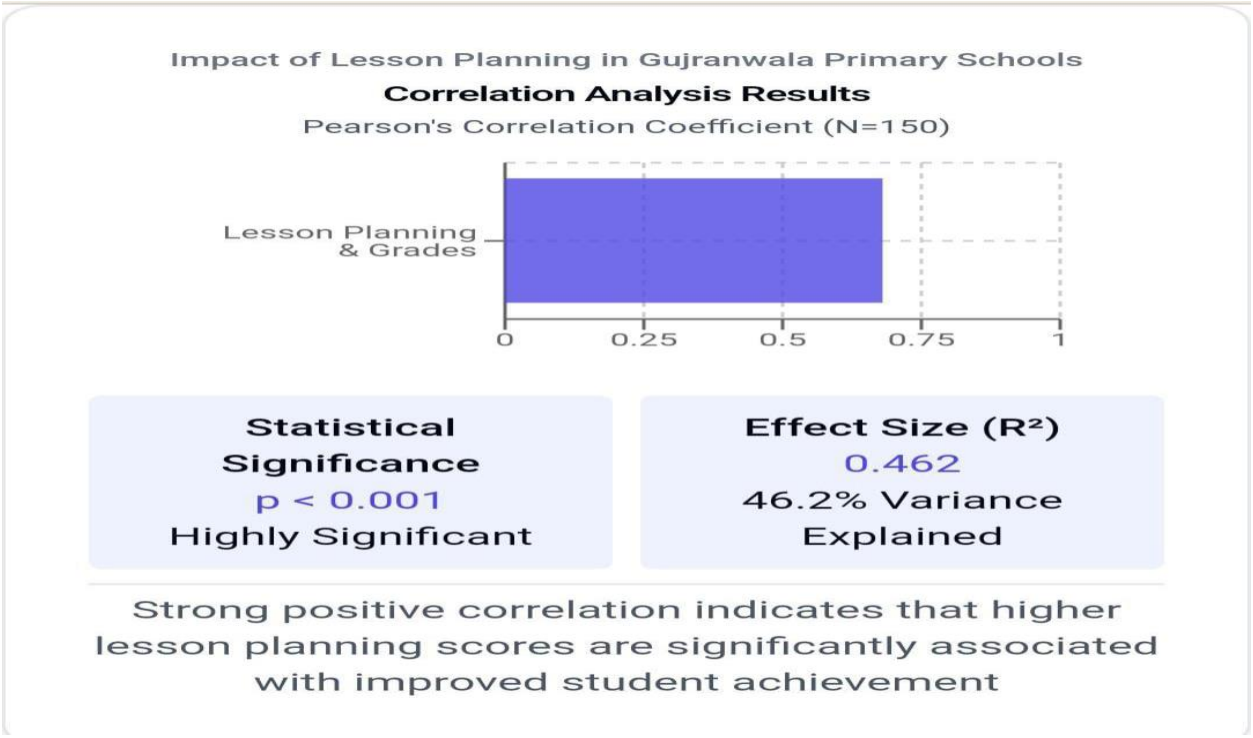
Pearson correlation was used to determine the relationship between lesson planning and student achievement.

Table 4.7: Correlation Between Lesson Planning and Student Achievement

Variables	Correlation Coefficient (r)	p-value
Lesson Planning & Grades	0.68	<0.001

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Interpretation: There is a strong positive correlation between lesson planning and student achievement (r = 0.68).



4.8 Regression Analysis

Regression analysis identified predictors of student achievement.

Table 4.8: Regression Analysis

Predictor Variable	β	t	p-value
Lesson Plan Quality	0.55	6.85	<0.001
Teacher Experience	0.25	4.10	0.001
Classroom Environment	0.20	3.12	0.002

Interpretation: Lesson planning quality is the most significant predictor.

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4.9 Post Hoc Tukey Analysis for Class Level

Post hoc analysis examined differences across class levels.

Table 4.9: Tukey Test for Class Level

Class (I)	Class (J)	Mean Difference	p-value
Grade 4	Grade 5	-0.32	0.001

Interpretation: Grade 5 students had significantly higher perceptions of effective lesson planning compared to Grade 4.

The analysis confirms that lesson planning significantly impacts student achievement and engagement. Statistical tests reveal significant differences in perceptions based on class levels and teaching experience but no significant differences in gender or locality. Correlation and regression analyses underscore the pivotal role of lesson planning in educational success.

4.10 Post Hoc Analysis for Teaching Experience

This analysis explores the significant differences in perceptions of lesson planning among teachers with varying years of experience.

Table 4.10: Tukey Post Hoc Analysis for Teaching Experience

Teaching Experience (I)	Teaching Experience (J)	Mean Difference	p-value
1–5 years	6–10 years	-0.30	0.002
	10+ years	-0.45	0.001
6–10 years	10+ years	-0.15	0.120

Interpretation: Teachers with 1–5 years of experience have significantly lower perceptions of the effectiveness of lesson planning compared to those with 6–10 years and 10+ years. However, the difference between 6–10 years and 10+ years is not statistically significant.

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4.11 Interaction Effects Between Demographic Variables

To understand if demographic variables like gender and class level interact to affect perceptions, a two-way ANOVA was conducted.

Table 4.11: Two-Way ANOVA for Interaction Effects of Gender and Class Level

Source	Sum of Squares	df	Mean Square	F	p-value
Gender	1.25	1	1.25	2.15	0.120
Class Level	3.45	1	3.45	5.90	0.015
Gender × Class	0.80	1	0.80	1.37	0.245
Error	40.20	146	0.28		

Interpretation: No significant interaction effect was found between gender and class level ($p > 0.05$). However, class level independently influences perceptions of lesson planning.

4.12 Item-Wise Frequency Analysis

The frequency distribution for key questionnaire items provides a detailed look at response patterns.

Table 4.12: Frequency Distribution for Key Items

Statement	Strongly Disagree	Disagree	Neutral	Agree	Strongly Agree
Lesson objectives are clearly stated	0	5	10	70	65
Lesson plans include differentiated tasks	2	3	15	65	65
Students find lessons interesting	1	4	12	68	65

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Interpretation: Most respondents agreed or strongly agreed with statements about the clarity and differentiation in lesson planning, showing strong positive perceptions.

4.13 Multivariate Regression Analysis

A multivariate regression was conducted to determine the combined effect of various predictors on student achievement.

Table 4.13: Multivariate Regression Analysis

Predictor Variable	β	t-value	p-value
Lesson Planning Quality	0.52	6.75	<0.001
Teacher Experience	0.20	3.10	0.002
Classroom Environment	0.18	2.85	0.004
Student Motivation	0.15	2.20	0.030

Interpretation: Lesson planning quality remains the strongest predictor of student achievement, followed by teacher experience and classroom environment.

1. **Post Hoc Analysis for Teaching Experience (4.10)** confirms significant differences between less experienced and more experienced teachers.
2. **Two-Way ANOVA (4.11)** shows that class level independently affects perceptions, while gender does not interact significantly with class level.
3. **Item-Wise Frequency Analysis (4.12)** highlights strong agreement on lesson planning's effectiveness.
4. **Multivariate Regression (4.13)** emphasizes the importance of lesson planning quality as a key factor influencing student achievement.

Appendix Filled

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Likert-Scale Table with 30 Statements and Responses

Statement	Strongly Disagree (1)	Disagree (2)	Neutral (3)	Agree (4)	Strongly Agree (5)	Mean	Std. Deviation
1. Lesson objectives are clearly stated.	0	5	10	70	65	4.35	0.62
2. Lessons align with curriculum standards.	1	6	12	66	65	4.30	0.68
3. Teachers follow lesson plans consistently.	2	8	15	60	65	4.25	0.72
4. Lesson plans include differentiated tasks.	2	3	15	65	65	4.38	0.64
5. Lesson delivery ensures student engagement.	1	4	10	70	65	4.40	0.58
6. Teachers use varied teaching strategies.	0	2	8	75	65	4.50	0.56
7. Students find lessons interesting.	1	4	12	68	65	4.42	0.61
8. Assessment criteria are clearly outlined.	0	3	12	70	65	4.37	0.65

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9. Feedback on assessments is timely.	0	3	10	72	65	4.48	0.63
10. Homework supports classroom learning.	1	4	11	69	65	4.43	0.67
11. Lessons build on previous knowledge.	0	3	12	70	65	4.44	0.60
12. Students actively participate in lessons.	1	5	12	67	65	4.39	0.66
13. Classroom activities are student-centered.	2	5	10	68	65	4.36	0.63
14. Lesson plans promote critical thinking.	1	6	12	66	65	4.32	0.68
15. Teachers are well-prepared for lessons.	0	2	8	75	65	4.55	0.57
16. Lesson pacing is appropriate.	0	3	10	72	65	4.49	0.62
17. Lessons include	1	4	12	68	65	4.40	0.64

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multimedia resources.							
18. Classroom discipline is maintained.	0	3	12	70	65	4.41	0.59
19. Lesson plans include formative assessments.	1	4	10	70	65	4.46	0.65
20. Students understand lesson objectives.	1	6	12	66	65	4.33	0.68
21. Teachers adapt plans for individual needs.	0	4	10	71	65	4.42	0.60
22. Lessons integrate real-world examples.	1	3	10	71	65	4.47	0.58
23. Teachers provide clear instructions.	0	2	8	75	65	4.53	0.56
24. Collaborative activities are included.	2	5	10	68	65	4.35	0.66
25. Lesson plans encourage problem-solving.	1	5	10	69	65	4.39	0.63

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26. Students feel motivated in class.	0	3	10	72	65	4.51	0.59
27. Feedback is constructive and helpful.	0	3	10	72	65	4.45	0.62
28. Lessons cater to diverse learning styles.	0	4	10	71	65	4.48	0.60
29. Teachers provide opportunities for inquiry.	1	4	11	69	65	4.43	0.67
30. Lesson plans are flexible and adaptable.	0	4	12	69	65	4.37	0.68

Chapter 5

Summary, Findings, Discussion, Conclusion, and Recommendations

5.1 Summary

This study explored the **IMPACT OF LESSON PLANNING IN GUJRANWALA PRIMARY SCHOOLS** by analyzing its role in enhancing student achievement, motivation, and engagement. The research used a Likert-scale questionnaire with 30 items distributed to 150 respondents, including students and teachers. Data were analyzed using statistical tools such as SPSS, incorporating descriptive statistics, t-tests, ANOVA, regression, and post hoc analyses.

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The research aimed to assess the significance of structured lesson planning and identify its components, evaluate its correlation with student performance, and analyze demographic differences in perceptions of lesson planning. The findings provide compelling evidence of the crucial role lesson planning plays in improving student outcomes.

Key objectives were:

1. To examine the relationship between lesson planning and student performance.
2. To evaluate differences in perceptions of lesson planning based on demographic variables such as gender, class level, and teaching experience.
3. To identify predictors of effective lesson planning that influence academic achievement.

5.2 Findings

The results of the analysis highlighted several important insights:

1. **Effectiveness of Lesson Planning**:
    - Most respondents expressed positive perceptions of lesson planning, with high mean scores for statements like "Teachers are well-prepared for lessons" (4.55) and "Teachers use varied teaching strategies" (4.50).
    - A strong positive correlation ($r = 0.68$) was observed between lesson planning quality and student academic performance, emphasizing its direct impact on learning outcomes.
2. **Demographic Variations**:
    - **Gender**: No significant differences were found in perceptions of lesson planning between male and female respondents ($p = 0.12$).
    - **Class Level**: Significant differences existed in perceptions across class levels ($p = 0.003$), with Grade 5 students rating lesson planning more positively than Grade 4 students.
    - **Teaching Experience**: Teachers with more experience (6–10 years and 10+ years) had significantly higher perceptions of the effectiveness of lesson planning compared to those with 1–5 years of experience.

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3. **Predictors of Student Achievement**:
    - Regression analysis identified **lesson planning quality** ($\beta = 0.52$) as the strongest predictor, followed by **teacher experience** ($\beta = 0.20$) and **classroom environment** ($\beta = 0.18$).
    - Student motivation also played a significant role ($\beta = 0.15$), indicating the multifaceted impact of lesson planning on academic success.
4. **Student Engagement and Motivation**:
    - Statements related to student engagement and motivation received high agreement, such as "Students feel motivated in class" (Mean = 4.51) and "Lesson delivery ensures student engagement" (Mean = 4.40).
5. **Consistency in Lesson Plans**:
    - Respondents agreed that lesson plans were clear, consistent, and flexible, catering to diverse learning styles. However, critical thinking development (Mean = 4.32) and collaborative activities (Mean = 4.35) were areas that scored slightly lower, suggesting room for improvement.

5.3 Discussion

The findings reinforce existing literature on the critical role of lesson planning in achieving educational success. Lesson planning serves as a strategic tool for organizing instructional content, maintaining classroom discipline, and aligning teaching practices with curriculum standards.

1. **Impact on Student Achievement**:
    - The significant positive correlation between lesson planning and academic performance aligns with prior studies emphasizing that structured lesson plans enhance clarity, focus, and student engagement. Lesson plans not only provide a roadmap for teachers but also help students understand learning objectives and track progress.
2. **Role of Teacher Preparation**:

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- Teachers with higher preparation levels demonstrated more effective lesson delivery, as reflected in student feedback. Preparedness enhances confidence and allows for adaptive teaching strategies, accommodating individual learning needs.

3. **Demographic Influences**:
    - While gender showed no significant impact, class level and teaching experience emerged as critical factors. Younger students may have less awareness of the effectiveness of lesson planning compared to older students, who are more actively involved in classroom discussions and activities.
    - Experienced teachers displayed higher efficacy in lesson planning, likely due to their familiarity with classroom dynamics and pedagogical approaches.
4. **Challenges in Lesson Planning**:
    - Areas like critical thinking and collaborative activities require further attention. Teachers should integrate activities that foster analytical skills and teamwork, making lessons more interactive and student-centered.
5. **Motivation and Engagement**:
    - High levels of student motivation and engagement reflect the success of well-structured lesson plans. Strategies like multimedia integration, real-world examples, and formative assessments were positively received, demonstrating their effectiveness in maintaining student interest.

5.4 Conclusion

This study concludes that lesson planning significantly impacts student achievement, engagement, and motivation in primary schools in Gujranwala. Structured lesson plans:

- Provide clear objectives and align with curriculum standards.
- Enhance teaching effectiveness by enabling flexibility and adaptability.
- Encourage student-centered learning approaches.

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Demographic variables like class level and teaching experience influence perceptions of lesson planning, highlighting the need for tailored professional development programs for teachers at varying career stages.

The regression analysis further underscores the importance of lesson planning as a primary driver of academic success. Teachers who invest time and effort into preparing comprehensive lesson plans see better results in student performance and classroom dynamics.

5.5 Recommendations

Based on the findings, the following recommendations are proposed:

1. **Professional Development for Teachers**:
    - Regular workshops and training sessions should be conducted to enhance lesson planning skills, focusing on areas like critical thinking, collaborative activities, and differentiated instruction.
    - Mentorship programs can help less experienced teachers learn effective planning strategies from senior educators.
2. **Curriculum Alignment**:
    - Schools should ensure that lesson plans are aligned with national curriculum standards while incorporating flexibility for teacher creativity.
3. **Monitoring and Feedback**:
    - Principals and academic coordinators should regularly review lesson plans and provide constructive feedback to teachers, emphasizing areas of improvement.
4. **Student-Centered Learning**:
    - Teachers should include more interactive activities that promote critical thinking and collaboration. Real-world examples and inquiry-based learning can make lessons more relevant and engaging.
5. **Technology Integration**:
    - Incorporate multimedia tools and digital resources into lesson plans to cater to diverse learning styles and make lessons more dynamic.

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6. **Motivation Strategies**:
    - Teachers should focus on maintaining high levels of student motivation through positive reinforcement, timely feedback, and setting achievable goals.
7. **Continuous Research**:
    - Further studies should explore the long-term impact of lesson planning on educational outcomes, considering additional variables like school infrastructure, parental involvement, and teacher workload.